

AMENDED CLAIMS

1. ~~A method for the treatment of mucositis in a patient comprising administering to the patient a therapeutically effective amount of at least one~~
5 ~~of an anthocyanoside, a proanthocyanidin, or an extract comprising at least one of an anthocyanoside or a proanthocyanidin for the treatment of mucositis.~~

2. ~~The method of claim 1, wherein the extract is derived from *Vaccinium myrtillus* or *Vitis vinifera*.~~

3. ~~The method of claim 1 further comprising administering a~~
10 ~~therapeutically effective amount of at least one of an anti-inflammatory agent, immunomodulating agent, analgesic, antimicrobial agent or antifungal agent.~~

4. ~~The method of claim 3, wherein the anti-inflammatory agent, immunomodulating agent or analgesic comprises an andrographolide or a natural or synthetic analogue thereof, a lactone sesquiterpene or a natural or~~
15 ~~synthetic analogue thereof, a parthenolide or a natural or synthetic analogue thereof, a cinnaropicrin or a natural or synthetic analogue thereof, a isobutylamide of a polyunsaturated fatty acid, or an extract of *Zanthoxylum bungeanum* or *Echinacea angustifolia*.~~

5. ~~The method of claim 3, wherein the antimicrobial agent or antifungal agent is natural or synthetic.~~
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6. ~~The method of claim 3, wherein the antimicrobial agent or antifungal agent comprises miconazole, an antibiotic, a benzofuran, an isoquinoline alkaloid, or an extract comprising miconazole, a benzofuran, or an isoquinoline alkaloid.~~

7. ~~The method of claim 3, wherein the antimicrobial agent or antifungal agent comprises a benzophenanthridine alkaloid or an extract comprising benzophenanthridine alkaloid.~~
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8. ~~The method of claim 3, wherein the antimicrobial agent or antifungal~~

~~agent comprises sanguinarine or a natural or synthetic analogue thereof; cheilerythrine or a natural or synthetic analogue thereof; cheilidenine or a natural or synthetic analogue thereof; eupomatoid or a natural or synthetic analogue thereof; or an extract of *Sanguinaria canadensis*, *Macleaya cordata* or *Macleaya microcarpa*.~~

9. ~~The method of claim 1, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is about 50 mg to about 500 mg per unit dose.~~

10. ~~The method of claim 1, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is about 3000 mg or less per day.~~

11. ~~The method of claim 1, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient by systemic administration.~~

15. ~~12. The method of claim 1, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient by oral or topical administration.~~

20. ~~13. The method of claim 3, wherein the therapeutically effective amount of the antimicrobial agent or the antifungal agent is administered to the patient by systemic administration.~~

~~14. The method of claim 3, wherein the therapeutically effective amount of the antimicrobial agent or the antifungal agent is administered to the patient by oral or topical administration.~~

25. ~~15. The method of claim 1, wherein the method comprises administering a pharmaceutical composition comprising (1) the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract, and (2) a pharmaceutically acceptable excipient.~~

~~16. The method of claim 15, wherein the pharmaceutical composition is in~~

the form of a tablet, a capsule, a chewing gum, a liquid form, a vaginal gel or granulates.

17. — The method of claim 15, wherein the pharmaceutical composition is in the form of a tablet and the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is contained in the tablet in an amount of about 10 mg to about 200 mg per tablet.

18. — The method of claim 15, wherein the pharmaceutical composition is in the form of a tablet and the tablet further comprises a therapeutically effective amount of a antimicrobial agent or antifungal agent in an amount of about 2 mg to about 10 mg per tablet.

19. — The method of claim 15, wherein the pharmaceutical composition is in the form of a slow dissolving tablet, and the extract comprises an extract of *Vaccinium myrtillus*, and wherein the pharmaceutical composition further comprises an isoquinoline alkaloid extracted from *Sanguinaria canadensis*, *Macleaya cordata* or *Macleaya microcarpa* and a lipophilic extract of *Zanthoxylum bungeanum*.

20. — The method of claim 1, wherein the mucositis occurs in a part of the patient's gastroenteric tract.

21. — The method of claim 1, wherein the mucositis occurs in the patient's mouth, esophagus, stomach, intestine, colon, sex organ or skin.

22. — The method of claim 1, wherein the mucositis occurs in the patient's mouth.

23. — The method of claim 1, wherein the patient has a cancer.

24. — The method of claim 1, wherein the mucositis occurs in connection with the administration of a chemotherapy or a radiotherapy to the patient.

25. — The method of claim 1, wherein the mucositis occurs in connection with the administration of a chemotherapy comprising the administration of an anthracycline, fluorouracil, paclitaxel, actinomycin, mithramycin, etoposide,

~~topotecan, amsacrine, methotrexate, hydroxyurea or a platinum complex.~~

~~26. The method of claim 1, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient before the patient receives a chemotherapy.~~

5 ~~27. The method of claim 1, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient while the patient receives a chemotherapy.~~

~~28. The method of claim 1, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to~~
10 ~~the patient after the patient receives a chemotherapy.~~

~~29. The method of claim 1, wherein the mucositis occurs in a sex organ of the patient and the method further comprises administering a therapeutically effective amount of a benzophenanthridine alkaloid, a benzofuran, a hyperforin or an analogue thereof, or a fluoroglucinol or an analogue thereof.~~

15 ~~30. A method for the prevention of mucositis in a patient comprising administering to the patient a therapeutically effective amount of at least one of an anthocyanoside, a proanthocyanidin, or an extract comprising at least one of an anthocyanoside or a proanthocyanidin for the prevention of mucositis.~~

20 ~~31. The method of claim 30, wherein the extract is derived from *Vaccinium myrtillus* or *Vitis vinifera*.~~

~~32. The method of claim 30 further comprising administering a therapeutically effective amount of at least one of an anti-inflammatory agent, immunomodulating agent, analgesic, antimicrobial agent or antifungal agent.~~

25 ~~33. The method of claim 32, wherein the anti-inflammatory agent, immunomodulating agent or analgesic comprises an andrographolide or a natural or synthetic analogue thereof, a lactone-sesquiterpene or a natural or synthetic analogue thereof, a parthenolide or a natural or synthetic analogue~~

thereof, a cynaropicrin or a natural or synthetic analogue thereof, a isobutylamide of a polyunsaturated fatty acid, or an extract of *Zanthoxylum bungeanum* or *Echinacea angustifolia*.

34. The method of claim 32, wherein the antimicrobial agent or antifungal agent is natural or synthetic.

35. The method of claim 32, wherein the antimicrobial agent or antifungal agent comprises miconazole, an antibiotic, a benzofuran, an isoquinoline alkaloid, or an extract comprising miconazole, a benzofuran, or an isoquinoline alkaloid.

36. The method of claim 32, wherein the antimicrobial agent or antifungal agent comprises a benzophenanthridine alkaloid or an extract comprising benzophenanthridine alkaloid.

37. The method of claim 32, wherein the antimicrobial agent or antifungal agent comprises sanguinarine or a natural or synthetic analogue thereof, chelerythrine or a natural or synthetic analogue thereof, chelidonium or a natural or synthetic analogue thereof, eupomatenoid or a natural or synthetic analogue thereof, or an extract of *Sanguinaria canadensis*, *Macleaya cordata* or *Macleaya microcarpa*.

38. The method of claim 30, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is about 50 mg to about 500 mg per unit dose.

39. The method of claim 30, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is about 3000 mg or less per day.

40. The method of claim 30, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient by systemic administration.

41. The method of claim 30, wherein the therapeutically effective amount

of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient by oral or topical administration.

42. — The method of claim 32, wherein the therapeutically effective amount of the antimicrobial agent or the antifungal agent is administered to the patient by systemic administration.

43. — The method of claim 32, wherein the therapeutically effective amount of the antimicrobial agent or the antifungal agent is administered to the patient by oral or topical administration.

44. — The method of claim 30, wherein the method comprises administering a pharmaceutical composition comprising (1) the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract, and (2) a pharmaceutically acceptable excipient.

45. — The method of claim 44, wherein the pharmaceutical composition is in the form of a tablet, a capsule, a chewing gum, a liquid form, a vaginal gel or granulates.

46. — The method of claim 44, wherein the pharmaceutical composition is in the form of a tablet and the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is contained in the tablet in an amount of about 10 mg to about 200 mg per tablet.

47. — The method of claim 44, wherein the pharmaceutical composition is in the form of a tablet and the tablet further comprises a therapeutically effective amount of a antimicrobial agent or antifungal agent in an amount of about 2 mg to about 10 mg per tablet.

48. — The method of claim 44, wherein the pharmaceutical composition is in the form of a slow dissolving tablet, and the extract comprises an extract of *Vaccinium myrtillus*, and wherein the pharmaceutical composition further comprises an isoquinoline alkaloid extracted from *Sanguinaria canadensis*, *Macleaya cordata* or *Macleaya microcarpa* and a lipophilic extract of

Zanthoxylum bungeanum.

49. — The method of claim 30, wherein the mucositis occurs in a part of the patient's gastroenteric tract.

50. — The method of claim 30, wherein the mucositis occurs in the patient's mouth, esophagus, stomach, intestine, colon, sex organ or skin.

51. — The method of claim 30, wherein the mucositis occurs in the patient's mouth.

52. — The method of claim 30, wherein the patient has a cancer.

53. — The method of claim 30, wherein the mucositis occurs in connection with the administration of a chemotherapy or a radiotherapy to the patient.

54. — The method of claim 30, wherein the mucositis occurs in connection with the administration of a chemotherapy comprising the administration of an anthracycline, fluorouracil, paclitaxel, actinomycin, mithramycin, etoposide, topotecan, ansacrine, methotrexate, hydroxyurea or a platinum complex.

55. — The method of claim 30, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient before the patient receives a chemotherapy.

56. — The method of claim 30, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient while the patient receives a chemotherapy.

57. — The method of claim 30, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is administered to the patient after the patient receives a chemotherapy.

58. — The method of claim 30, wherein the mucositis occurs in a sex organ of the patient and the method further comprises administering a therapeutically effective amount of a benzophenanthridine alkaloid, a benzofuran, a hyperforin or an analogue thereof, or a flaveroglucinol or an analogue thereof.

59. A pharmaceutical composition for use in the prevention or treatment or

prevention of mucositis comprising (1) a therapeutically effective amount of at least one of an anthocyanoside, a proanthocyanidin, or an extract comprising at least one of an anthocyanoside or a proanthocyanidin, and (2) a pharmaceutically acceptable excipient.

- 5 60. The composition of claim 59, wherein the extract is derived from *Vaccinium myrtillus* or *Vitis vinifera*;
61. The composition of claim 59 further comprising a therapeutically effective amount of at least one of an anti-inflammatory agent, immunomodulating agent, analgesic, antimicrobial agent or antifungal agent.
- 10 62. The composition of claim 61, wherein the anti-inflammatory agent, immunomodulating agent or analgesic comprises an andrographolide or a natural or synthetic analogue thereof, a lactone sesquiterpene or a natural or synthetic analogue thereof, a parthenolide or a natural or synthetic analogue thereof, a cynaropicrin or a natural or synthetic analogue thereof, a
- 15 isobutylamide of a polyunsaturated fatty acid, or an lipophilic extract of *Zanthoxylum bungeanum* or *Echinacea angustifolia*, and
63. The composition of claim 61, wherein the antimicrobial agent or antifungal agent is natural or synthetic;
64. The composition of claim 61, wherein the antimicrobial agent or
- 20 antifungal agent comprises miconazole, an antibiotic, a benzofuran, an isoquinoline alkaloid, or an extract comprising miconazole, a benzofuran, or an isoquinoline alkaloid;
65. The composition of claim 61, wherein the antimicrobial agent or antifungal agent comprises a benzophenanthridine alkaloid or an extract
- 25 comprising benzophenanthridine alkaloid;
66. The composition of claim 61, wherein the antimicrobial agent or antifungal agent comprises sanguinarine or a natural or synthetic analogue thereof, chelerythrine or a natural or synthetic analogue thereof, chelidamine

or a natural or synthetic analogue thereof, eupomatenoid or a natural or synthetic analogue thereof, or an extract of *Sanguinaria canadensis*, *Macleaya cordata* or *Macleaya microcarpa*.

67. ~~The composition of claim 59, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is about 50 mg to about 500 mg per unit dose.~~

68. ~~The composition of claim 59, wherein the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is about 3000 mg or less per day.~~

10 269. The A pharmaceutical composition for use according to claim 159, wherein the composition is suitable for systemic administration.

370. The A pharmaceutical composition for use according to claim 159, wherein composition is suitable for oral or topical administration.

15 471. The A pharmaceutical composition for use according to claim 159, wherein the composition is in the form of a tablet, a capsule, a chewing gum, a liquid form, a vaginal gel or granulates.

72. ~~The composition of claim 59, wherein the composition is a non-oral composition and the pharmaceutically acceptable excipient is suitable for non-oral administration.~~

20 73. ~~The composition of claim 59, wherein the composition is in the form of a tablet and the therapeutically effective amount of the at least one anthocyanoside, proanthocyanidin or extract is contained in the tablet in an amount of about 10 mg to about 200 mg per tablet.~~

25 74. ~~The composition of claim 59, wherein the composition is in the form of a tablet and the tablet further comprises a therapeutically effective amount of an antimicrobial agent or antifungal agent in an amount of about 2 mg to about 10 mg per tablet.~~

75. ~~The composition of claim 59, wherein the composition is in the form of~~

~~a slow dissolving tablet, and the extract comprises an extract of *Vaccinium myrtillus*, and wherein the composition further comprises an isoquinoline alkaloid extracted from *Sanguinaria canadensis*, *Macleaya cordata* or *Macleaya microcarpa* and a lipophilic extract of *Zanthoxylum bungeanum*.~~

- 5 5. A pharmaceutical composition for use according to claims 1-4, wherein the mucositis occurs in a part of the patient's gastroenteric tract.
6. A pharmaceutical composition for use according to claims 1-4, wherein the mucositis occurs in the patient's mouth, esophagus, stomach, intestine, colon, sex organ or skin.
- 10 7. A pharmaceutical composition for use according to claims 1-4, wherein the mucositis occurs in the patient's mouth.
8. A pharmaceutical composition for use according to claims 1-4, wherein the patient has a cancer.
9. A pharmaceutical composition for use according to claims 1-4, wherein
15 the mucositis occurs in connection with the administration of a chemotherapy or a radiotherapy to the patient.
10. A pharmaceutical composition for use according to claims 1-4, wherein the mucositis occurs in connection with the administration of a chemotherapy comprising the administration of an anthracycline, fluorouracyl, paclitaxel,
20 actinomycin, mithramycin, etoposide, topotecan, amsacrine, methotrexate, hydroxyurea or a platinum complex.
11. A pharmaceutical composition for use according to claim 1, wherein said composition is administered to the patient before, while or after the
25 patient receives a chemotherapy.